ATOMIC STRUCTURE — From Thomson's Plum Pudding Model to Modern Quantum Theory

■ A group of 10 students teaches this story step by step: how the electron was found, how Thomson built a model, how Rutherford broke it, and why even Rutherford had to be replaced. ■ Driving question for the whole hour: if atoms contain charged parts, what holds them together — and why do atoms not collapse? ■ You will see real dates, numbers and equations on every slide, so you can follow and verify each claim from your seat. ■ Team: Members 01–10 (names TBD — numbered placeholders). College: [College Name TBD]. Presenter order follows the number on each slide. ■ How to use this deck: read each slide's speaking points aloud in your own words, point at the

Visual: hero: navy cover, amber nucleus + 3 cyan/turquoise/violet orbit rings, electrons as dots; caption warns rings are historical orbits corrected on Slide 19.

Our Learning Journey — Five Models, One Method: Experiment Corrects Theory

■ Science advances when a new experiment breaks the old picture: Dalton's solid ball → Thomson's pudding → Rutherford's nucleus → Bohr's orbits → quantum orbitals. ■ Member 1 opens with Dalton and the three subatomic particles; Member 2 shows how cathode rays revealed the electron in 1897. ■ Members 3–4 present Thomson's 1904 model and why it failed; Members 5–7 rebuild the atom from Rutherford's 1909 gold-foil evidence. ■ Members 8–9 compare the models and show why classical physics predicts collapse in ~10■¹¹ s and a wrong continuous spectrum. ■ Member 10 closes the arc: Bohr's 1913 quantised orbits, then Heisenberg–Schrödinger orbitals — with

Visual: ivory card: horizontal 5-node timeline 1808→1897→1911→1913→1926 with Thomson/Rutherford nodes highlighted amber.

What Is an Atom? From 'Atomos' to Dalton's Scientific Theory (1808)

■ The word atom comes from Greek atomos, meaning indivisible — an old philosophical idea that Dalton turned into testable chemistry in 1808. ■ Dalton proposed: matter is made of tiny atoms; atoms of one element are identical; atoms combine in fixed ratios to form compounds — pictured as a tiny solid ball that can never be broken. ■ Dalton's picture explained combining ratios chemists could weigh, so it survived nearly a century despite having no internal structure. ■ Later, electrolysis and electric discharge through gases showed atoms are connected to electricity — they must contain smaller charged parts, so atoms are divisible. ■ That forces our central question, which

Visual: ivory card: 5 Dalton solid-ball spheres + CO vs CO2 fixed-ratio mass bars; caption on weighable combining ratios.

The Three Subatomic Particles — Electron, Proton, Neutron

■ The electron carries –1.6 × 10■[1]■ C with a tiny mass of ~9.1 × 10■[31] kg — the lightest particle, first proof that atoms can be broken apart. ■ The proton carries the same charge but positive (+1.6 × 10■[1]■ C) and is ~1836 times heavier than the electron — it anchors the atom's positive charge. ■ The neutron, found later by Chadwick in 1932, has no charge and a mass almost equal to the proton — it completes the trio but post-dates Thomson and Rutherford. ■ Together they explain neutrality: total positive charge equals total negative charge, so the whole atom is electrically neutral overall. ■ TABLE (right): read charge, mass and discoverer row by row — we spend the rest of

Visual: native editable table 4x4: e-/p+/n0 charge/mass/discoverer (Thomson 1897, Chadwick 1932); neutrality rule caption.

Discovery of the Electron (1897) — The Cathode-Ray Tube Apparatus

■ J. J. Thomson discovered the electron in 1897 while studying cathode rays — streams from cathode to anode in a sealed glass tube at very low pressure under high voltage. ■ Apparatus (diagram right): evacuated glass tube, cathode (–) and anode (+) electrodes, high-voltage supply, fluorescent screen that glows where rays strike. ■ With no fields applied, rays travel straight from cathode to anode — the baseline path before we test what they are. ■ Low pressure matters: too much gas would scatter the rays off air molecules; vacuum lets the beam fly cleanly to the screen. ■ Portrait: J. J. Thomson (photo, public domain) — his 1897 deflection measurements, not the glow alone,

Visual: custom CRT schematic (tube, cathode/anode, cyan beam, amber screen glow) + embedded J.J. Thomson portrait (PD, pre-1915).

Properties of Cathode Rays — Why They Must Be Negative Particles

■ Cathode rays travel in straight lines: they cast a sharp shadow of any object placed in their path — light-like straightness, but they are not light. ■ They carry energy and momentum: they spin a tiny paddle wheel placed in the beam and make glass or a fluorescent screen glow where they hit. ■ Crucial test — in an electric field they bend toward the positive plate, proving they carry negative charge; magnetic fields deflect them likewise. ■ Thomson balanced electric against magnetic deflection until they cancelled, then computed e/m = 1.76 × 10^{11} C/kg — ~1800× larger than a hydrogen ion, so the particle is extremely light. ■ The same e/m appeared whatever gas or

Visual: custom deflection schematic: red + plate top, navy - plate bottom, cyan beam bending upward + embedded modern CRT photo (Hebels, PD, illustrative).

Thomson's Plum Pudding Model (1904) — A Sphere of Positive Charge Studded with Electrons

■ Question Thomson faced: electrons are negative but atoms are neutral — where is the positive charge? His 1904 answer: spread uniformly through a sphere ~10■[1]■ m in radius. ■ Picture (right): the 'pudding' is diffuse positive charge; embedded 'plums' are electrons — also called the watermelon model: red pulp = positive, seeds = electrons. ■ Key rule: total positive charge equals total negative charge, so the atom is electrically neutral overall — a success of the model. ■ There is NO nucleus in this model — positive charge fills the whole atom, and electrons sit or vibrate about fixed positions inside it. ■ Thomson even used vibrating electrons to try to explain light emission — a

Visual: custom pudding: amber uniform sphere R~1e-10 m with navy electron dots; watermelon-twin caption.

Why Thomson's Model Failed — Four Problems Before a Single Gold-Foil Shot

■ Large-angle scattering impossible: spread-out positive charge gives only a weak internal field, predicting tiny deflections — yet experiments later showed violent rebounds. ■ Electrons cannot rescue it: an alpha particle is ~7300× heavier than an electron — like a bus hitting ping-pong balls, the bus will not bounce back. ■ No nucleus, no concentration: the model assumes positive charge fills the atom, leaving no dense centre that could throw back a fast heavy particle. ■ Wrong light and fragile stability: vibrating electrons could not give sharp line spectra at exact wavelengths, and static charges cannot balance stably by electrostatics alone. ■ Verdict: the model explained

Visual: red-bordered card: 'only tiny nudges' prediction, diffuse atom with straight cyan pass + bus-vs-ping-pong mass analogy (7300x).

Rutherford's Gold-Foil Experiment (1909, Manchester) — Setup That Broke the Pudding

■ Aim (Rutherford, Geiger & Marsden, ~1909, Manchester): fire fast heavy particles at atoms and watch deflections — a direct test of Thomson's spread-charge prediction. ■ Alpha source: radioactive radium/polonium emitting helium nuclei — charge +2e, speed ~10■ m/s — housed in a lead block with a narrow hole that makes a thin beam (lead absorbs the rest). ■ Target: very thin gold foil, only a few hundred atoms thick — gold chosen because it hammers extremely thin and its heavy nucleus (Z = +79) gives strong repulsion. ■ Detector: zinc-sulphide screen around the foil flashes (scintillation) wherever an alpha hits; a microscope counts flashes angle by angle — whole

Visual: dark card top-view schematic: Pb collimator, thin Au foil (Z=79), cyan transmitted/bent + red rebound paths, Rutherford portrait (PD) + historic Geiger-Marsden thumbnail (PD, redrawn).

Three Observations — What Happened vs What It Meant (Keep Them Separate)

■ OBS. 1 — Most alphas passed straight through with zero deflection. CONCLUSION: the atom is mostly empty space — the beam usually meets nothing. ■ OBS. 2 — Some alphas deflected by small angles (~0.14% by more than 1°). CONCLUSION: concentrated positive charge exists inside and repels the +2e alphas. ■ OBS. 3 — Very few, ~1 in 8000, scattered by >90°, some straight back at 180°. CONCLUSION: positive charge + almost all mass sit in a tiny dense nucleus — only a head-on hit can reverse a fast heavy alpha. ■ Rutherford's image: like firing a huge shell at tissue paper and watching it come back — the most surprising result, impossible for spread charge. ■

Visual: dark card: foil atom dots with 3 paths — straight cyan, small-bend cyan, 180-degree red rebound; header stats 0.14% / 1 in 8000.

Impact Parameter — Why Aim Decides Deflection

■ Definition: impact parameter b = perpendicular distance between the alpha's original straight path and the nucleus centre — how 'off-centre' the aim is. ■ Large b (wide miss): alpha feels only weak distant repulsion and sails on with a tiny deflection — the common case, hence mostly straight-through. ■ Small b (close pass): alpha dives through the strong near-nucleus field and swings through a large angle — the rare ~0.14% tail. ■ Zero b (head-on, b = 0): alpha stops, turns and bounces back at 180° — the ultra-rare ~1 in 8000 events that killed Thomson's model. ■ Rule to memorize: smaller impact parameter → closer approach → larger deflection; draw the three paths

Visual: custom 3-path diagram around amber nucleus with b-bracket: large b tiny bend, small b large bend, b=0 180-degree back.

Rutherford's Nuclear Model (1911) — Tiny Dense Nucleus + Orbiting Electrons

■ Core claim (1911): a small, dense, positively charged nucleus sits at the centre holding most of the atom's mass; electrons revolve around it in circular orbits. ■ Mostly empty space: the nucleus is ~10■–10■× smaller than the atom, so most alphas never approach one — straight-through explained at once. ■ Also called the planetary model: electrons orbit like planets, with electrostatic attraction playing the role gravity plays for planets. ■ Force picture (historical): Coulomb attraction provides centripetal force — F(Coulomb) = m·v²/r — nucleus pulls in, circular motion needs inward force. ■ Honest label: this orbit picture is HISTORICAL — it explains scattering beautifully but keeps

Visual: dark card: amber nucleus + 2 cyan/turquoise orbit rings + electron dots; centered equation F(Coulomb) = m v^2/r with historical-orbit warning.

How Small Is the Nucleus? The Stadium-and-Seed Scale

■ Source sizes: nucleus radius ~10■¹■ to 10■¹■ m; whole-atom radius ~10■¹■ m — the atom is ~10,000 to 100,000× wider than its nucleus. ■ Analogy from the script: if the atom were a football stadium, the nucleus would be a tiny seed at the centre — everything else is empty space with light electrons. ■ Bar (right, log scale): dot for nucleus vs long bar for atom — you cannot draw both true-to-scale on one linear diagram, so we use a magnifier inset. ■ Why scattering looks so empty: nuclei are rare tiny targets; only a direct hit samples the strong near-field — hence 1 in 8000 large rebounds. ■ Keep these two numbers (10■¹■ m vs 10■¹■–10■¹■ m) — the teacher will ask them,

Visual: log-scale card: long cyan atom bar (1e-10 m) vs magnified amber nucleus dot (1e-15–1e-14 m) + stadium:seed band.

Thomson vs Rutherford — Same Electrons, Opposite Architecture

■ One-line contrast: Thomson imagined a uniform FILLED atom; Rutherford proved a mostly EMPTY atom with a tiny dense centre — positive-charge placement is the whole difference. ■ TABLE (right): Positive charge — spread through atom vs concentrated in nucleus; Electrons — embedded vs revolving; Nucleus — absent vs present. ■ Row 4 of table: Alpha scattering — Thomson failed to explain it vs Rutherford explained it (strong near-field only near nucleus). ■ Credit both: Thomson proved electrons live inside atoms; Rutherford discovered the nucleus — each step used new experimental evidence to correct the last. ■ Bridge: both are still CLASSICAL models — and

Visual: native editable 5x3 comparison table (charge/electrons/nucleus/scattering) + bridge note both classical.

Why Classical Orbits Collapse — The ~10^-11 s Death Sentence (a Prediction)

■ Classical rule (Maxwell): any accelerated charge radiates electromagnetic waves and loses energy — circular motion IS acceleration because direction keeps changing. ■ Apply to Rutherford's electron: it should radiate continuously, lose energy, spiral inward with shrinking radius — and crash into the nucleus. ■ Timescale from the source: collapse in only ~10■^{11} s — labelled clearly as a CLASSICAL PREDICTION, not a measurement; real atoms are stable, so the prediction refutes the theory. ■ Diagram (right): healthy circle → decaying spiral → crash, with wavy radiation arrows bleeding energy at every turn. ■ Upshot: stability alone kills the classical orbit — and the

Visual: red-bordered card: amber nucleus + decaying cyan-to-red spiral + radiation squiggles; collapse labeled PREDICTION.

Continuous vs Line Spectra — The Fingerprint Classical Physics Cannot Print

■ Classical expectation: a spiralling electron's frequency changes continuously, so it should emit ALL wavelengths — a smooth rainbow continuum. ■ Reality: hydrogen and other atoms emit only SHARP lines at specific wavelengths — e.g. hydrogen's visible Balmer series (red, cyan-blue, violet lines on the photo right). ■ Those lines follow the Rydberg formula (supplementary: 1/λ = R(1/n■² – 1/n■²), R ≈ 1.097×10■ m■¹) — exact integer pattern Rutherford's spirals cannot give. ■ So two independent failures: atoms do NOT collapse (stable) and do NOT glow continuously (line spectra) — classical physics fails at atomic scale. ■ A new theory must quantise something — Member

Visual: navy continuum band vs embedded H emission photo (Umop503 CC0, Balmer lines) + Rydberg formula note (supplementary, R=1.097e7 m^-1).

Bohr's Fix (1913) — Stationary Orbits, Quantised Angular Momentum, Quantum Jumps

■ Postulate 1 — Stationary orbits: electrons may circle only in certain allowed orbits and radiate NO energy while in them — stability imposed by rule. ■ Postulate 2 — Quantised angular momentum: only orbits with L = n·h/2π, n = 1, 2, 3… are allowed — orbits are quantised in whole steps, nothing between. ■ Postulate 3 — Quantum jumps: light is emitted/absorbed ONLY when an electron jumps between orbits, carrying ΔE = h·ν — the energy difference, not a spiral glow. ■ Payoff: jumps between fixed levels give fixed photon energies → sharp lines — hydrogen's spectrum explained for the first time. ■ Portrait: Niels Bohr c.1922 (public domain, Nobel year) — his model

Visual: 3 postulate blocks + Bohr c.1922 portrait (PD); central equations L = n h/2pi and DeltaE = h nu.

Bohr Energy Levels — The Ladder Electrons Climb and Fall (Hydrogen)

■ Hydrogen ladder (source formula): E(n) = –13.6/n² eV — n = 1 lowest (–13.6 eV), n = 2 (–3.4 eV), n = 3 (–1.51 eV), n = 4 (–0.85 eV), … up to 0 (free). ■ Diagram (right): horizontal rungs n = 1–4 with gaps shrinking upward — longer UP arrow = absorption (photon in, electron climbs); longer DOWN arrow = emission (photon out). ■ Photon rule on every arrow: ΔE = h·ν — big jump = bluer, more energetic light; small jump = redder light — hence discrete Balmer lines. ■ Worked check: 2→1 fall releases ~10.2 eV (ultraviolet Lyman); Balmer falls to n = 2 give visible lines — integers in, wavelengths out. ■ Scope warning (source): Bohr fits hydrogen-like atoms only — heavier atoms

Visual: custom ladder n=1..4 (-13.6/-3.4/-1.51/-0.85 eV) with violet absorb-up and amber emit-down arrows; En=-13.6/n^2 eV header.

Modern Quantum Model — From Fixed Orbits to Probability Orbitals

■ Heisenberg (1927, supplementary context): we cannot know an electron's exact path and speed together — exact classical trajectories are unknowable in principle. ■ Schrödinger's wave equation (1926) replaces paths with wavefunctions: |ψ|² gives the PROBABILITY of finding the electron in each region of space. ■ An orbital is that region — e.g. spherical 1s cloud densest at centre (image right: brighter = likelier; PoorLeno, public domain) — described by quantum numbers n, l, m, s. ■ ORBIT vs ORBITAL (exam trap): Bohr orbit = fixed circular path with exact radius; quantum orbital = fuzzy probability cloud with no exact path — never swap the words. ■ Result:

Visual: embedded H density plots (PoorLeno PD, brighter=likelier) + ORBIT vs ORBITAL contrast blocks; quantum numbers n,l,m,s.

Evolution Timeline — Dalton → Thomson → Rutherford → Bohr → Quantum

■ 1808 Dalton: solid indivisible ball — explains combining ratios, no inner parts; broken by cathode-ray electricity evidence. ■ 1897–1904 Thomson: electron found (e/m = 1.76×10¹¹ C/kg) → plum-pudding sphere ~10■¹■ m, neutral but nucleus-free; broken by large-angle scattering. ■ 1909–1911 Rutherford/Geiger/Marsden: 1 in 8000 rebound → tiny dense nucleus (10■¹■–10■¹■ m), planetary orbits; broken by collapse + line spectra. ■ 1913 Bohr: quantised orbits L = n·h/2π, jumps ΔE = hν, En = –13.6/n² eV — fixes hydrogen lines; limited to hydrogen-like atoms. ■ 1926–present Quantum: Heisenberg + Schrödinger → probability orbitals (n, l, m, s) — current teaching model; each

Visual: dark card 5-node timeline with breaker experiment per step; method caption.

Quick Revision — Eight Questions to Lock the Story

■ Q1: Who found the electron, when? — Thomson, 1897, via cathode rays with e/m = 1.76×10¹¹ C/kg, same for all gases/metals. ■ Q2: What IS Thomson's model? — Positive sphere ~10■¹■ m with embedded electrons, neutral, NO nucleus (pudding/watermelon). ■ Q3: What was Rutherford's experiment? — 1909 gold-foil alpha scattering: +2e alphas at ~10■ m/s onto thin gold (Z=79), ZnS flashes in vacuum. ■ Q4: Three results? — Most straight (empty space); ~0.14% small angles (positive charge); ~1 in 8000 >90°/back (tiny dense nucleus). ■ Q5: Rutherford's model + scale? — Dense central nucleus + orbiting electrons, mostly empty; nucleus 10■¹■–10■¹■

Visual: 8 numbered navy chips with one-line cues (Q1 Thomson 1897 … Q8 orbit vs orbital); answers are slide-left bullets.

Teacher Cross-Questions (1–5) — Answered by the Topic Owner

■ Q1 [MEMBER 01]: What is an atom? — Smallest particle of an element taking part in chemical reactions; once thought indivisible, now known to hold electrons, protons, neutrons. ■ Q2 [MEMBER 03]: Why 'plum pudding'? — Electrons embedded in a positive sphere like plums in pudding — or seeds in watermelon pulp (red pulp = +, seeds = e■). ■ Q3 [MEMBER 02]: What are cathode rays? — Streams of negative particles (electrons) flying cathode→anode in a low-pressure tube; deflected by electric/magnetic fields. ■ Q4 [MEMBER 04]: Why did Thomson's model fail? — No nucleus, cannot give large-angle scattering, cannot give exact line spectra or stable

Visual: dark Q&A card: owner-stands rule + per-question member tags M01/M03/M02/M04/M05 chips.

Teacher Cross-Questions (6–10) — Answered by the Topic Owner

■ Q6 [MEMBER 06]: Why did most alphas pass straight? — The atom is mostly empty space; most alphas never came near any nucleus (large impact parameter). ■ Q7 [MEMBER 06]: Why were a few greatly deflected? — They passed very close to the tiny dense + nucleus and were violently repelled; small b → large angle, b = 0 → 180° back. ■ Q8 [MEMBER 07]: What is Rutherford's nuclear model? — Small dense positive nucleus holds most mass; electrons revolve around it; rest is empty space. ■ Q9 [MEMBER 07]: Why 'planetary'? — Electrons circle the nucleus like planets circle the Sun, with electrostatic attraction replacing gravity ($F = mv^2/r$). ■ Q10 [MEMBER 09]:

Visual: dark Q&A card: tags M06/M06/M07/M07/M09; impact-parameter and planetary-gravity points.

Conclusion — From Solid Ball to Probability Cloud

■ Dalton (1808) gave us atoms; Thomson (1897/1904) gave us the electron and the first inner model — neutral but nucleus-free and soon broken. ■ Rutherford (1909–1911) gave us the nucleus and the planetary picture — tiny dense centre (10■¹■–10■¹■ m) in vast emptiness (10■¹■ m) — broken in turn by stability and spectra. ■ Bohr (1913) quantised the orbits (L = n·h/2π, ΔE = hν, En = –13.6/n² eV) and saved hydrogen's lines; quantum theory (Heisenberg–Schrödinger) replaced orbits with orbitals. ■ Method to remember: experiment breaks theory, numbers decide — e/m, 1 in 8000, 10■¹■ m, 10■¹¹ s, –13.6 eV are the fingerprints of each break. ■ “Respected teacher and

Visual: dark card: SOLID BALL→PUDDING→NUCLEUS→ORBITS→CLOUDS arc + verbatim source closing quote in amber.

References, Sources & Image Credits — Trace Every Claim and Picture

■ Teaching source (primary): 'Atomic Structure — Detailed Teach-the-Teacher Script' (8 pp., 10-member script) — all dates, numbers, equations and 10 Q&A above come from it. ■ Supplementary science (standard values, flagged in notes): Balmer/Rydberg context, Bohr constants, Heisenberg 1927 / Schrödinger 1926 framing — verify against your course textbook. ■ Historic images (Wikimedia Commons, accessed 2026-10-15): J.J. Thomson pre-1915 photo (PD-old); Rutherford — Bain News Service via LoC LCCN 2014716719, rest. Bammesk (PD-US); Bohr c.1922 AB Lagrelius & Westphal (PD); Geiger–Marsden 1913 apparatus diagram (PD, low-res — redrawn on slides);

■ PRIMARY: Teach-the-Teacher Script (8 pp.) — all dates/numbers/Q&A. ■ SUPPL: Rydberg R, Bohr constants, 1926–27 quantum framing — check textbook. ■ IMAGES: Commons PD/CC0 (see manifest) — Thomson / Rutherford (LoC) / Bohr / Geiger–Marsden / Umop503 / PoorLeno / Hebels. ■ DIAGRAMS: custom PPT schematics — not to scale unless stated. ■ MANIFEST: work/image_asset_manifest.csv • Accessed 2026-10-15

Visual: dark credits card: primary script source, supplementary flag, per-image PD/CC0 credits, manifest pointer.